Assignment - 2

1. Determine the bionamial distribution for which the mean is 4 and variance is 3.

Sol

Given mean $(np) = 4$

variance $(npq) = 3$.

$npq = 3$

$4(q) = 3$

$q = 3/4$

w.k.t $p + q = 1$

$p = 1 - q$

$p = 1 - 3/4 = 1/4$

$np = 4$

$n = 4/p = 16$

$P(X = r) = {}^{n}C_r (p)^r (q)^{n-r}$

$= {}^{16}C_r (1/4)^r (3/4)^{16-r}$

where $r = 0, 1, \ldots, 16$.

2. Out of 800 families with 5 children each how many would you expect to have (a) 3 boys (b) 5 girls (c) either 2 (or) 3 boys (d) at least one boy (Assume equal probabilities for boys and girls).

Sol

Given Number of families = 800

Number of children in each family $n = 5$

Probability of a boy $p = 1/2$

Probability of a girl $q = 1/2$

$$P(X = x) = {}^{n}C_x (p)^x (q)^{n-x}$$

i, 3 boys

$$P(X = 3) = {}^{5}C_3 (1/2)^3 (1/2)^2$$

$$= {}^{5}C_3 \left(\frac{1}{2}\right)^5$$

$$= \frac{5!}{3!2!} \times \frac{1}{32}$$

$$= \frac{5 \times 4 \times 3}{3 \times 2} \times \frac{1}{32_8} = \frac{5}{16}$$

Expected families : $800 \times \frac{5}{16} = 250.$

ii, 5 girls :

5 girls means 0 boys.

$$P(X = 0) = {}^{5}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5$$

$$= \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Expected families $800 \times \frac{1}{32} = 25$

iii, either 2 or 3 boys

$$P(X = 2) = {}^{5}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = {}^{5}C_2 \left(\frac{1}{2}\right)^5 = \frac{5}{16}$$

$$P(X = 3) = P(X = 2) = \frac{5}{16}$$

$$P(2 \text{ or } 3) = \frac{5+5}{16} = \frac{10}{16} = \frac{5}{8}$$

Expected families : $800 \times \frac{5}{8} = 500$

(d) families having atleast one boy.

$$P(\text{atleast one boy}) = 1 - P(\text{no boys})$$
$$= 1 - P(X=0)$$
$$= 1 - {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5$$
$$= 1 - \frac{1}{25} = 1 - \frac{1}{32}$$
$$= \frac{31}{32}$$

Expected families :

$$\Rightarrow 800 \times \frac{31}{32} = 775$$

3. The average number of homes sold by the Realty Company is 2 homes per day. Determine the probability that will be sold tomorrow. i, exactly 3 homes ii, in b/w 0 and 2, both inclusive.

Sol Given, Mean number of homes sold per day

$$\lambda = 2$$

Poisson probability formula

$$P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

i, probability of selling exactly 3 homes

$$k = 3, \quad \lambda = 2$$

$$P(X=3) = \frac{e^{-2}(2)^3}{3!} \qquad [e^{-2} = 0.1353]$$
$$= \frac{e^{-2} \times 8}{6}$$
$$= \frac{1.0824}{6} = 0.1804$$

ii) probability of selling b/w 0 and 2 homies

$$P(0 \le x \le 2) = p(x=0) + P(x=1) + p(x=2)$$

$\Rightarrow$ At $x=0 \Rightarrow P(x=0) = \frac{e^{-2}(2)^0}{0!} = e^{-2} = 0.1353$

$\Rightarrow$ At $x=1 \Rightarrow P(X=1) = \frac{e^{-2}(2)^1}{1!} = 2(0.1353)$

$= 0.2706$

$\Rightarrow$ At $x=2 \Rightarrow P(X=2) = \frac{e^{-2}(2)^2}{2!}$

$= \frac{4(0.1353)}{2} = 0.2706$

$P(0 \le x \le 2) = 0.1353 + 0.2706 + 0.2706$

$= 0.6765$

4. The average number of phone calls per minute coming into a switch board b/w 2 p.m and 4 p.m is 2.5. Determine the probability that during one particular minute there will be i) 4 (or) fewer ii) more than 6 calls.

Sol

Given Average Number of phone calls per minute $(\lambda) = 2.5$

Poisson probability :

$$P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

i) probability of 4 (or) fewer calls.

$$P(X \le 4) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4)$$

At $k=0$, $P(X=0) \Rightarrow \frac{e^{-2.5}(2.5)^0}{0!} = 0.0821$

$[e^{-2.5} = 0.0821]$

At $k=1$, $P(X=1) \Rightarrow \frac{e^{-2.5}(2.5)^1}{1!} = 0.2052$

At $k=2$, $P(X=2) \Rightarrow \frac{e^{-2.5}(2.5)^2}{2!} = 0.2565$

At $k=3$, $P(X=3) \Rightarrow \frac{0.0821(2.5)^3}{3!} = 0.2138$

At $k=4$, $P(X=4) \Rightarrow \frac{0.0821(2.5)^4}{4!} = 0.1336$

$P(X \leq 4) = 0.0821 + 0.2052 + 0.2565 + 0.2138 + 0.1336$

$= 0.8912$

(ii) probability of more than 6 cells.

$P(X \geq 6) = 1 - P(X \leq 6) = 1 - P(X \leq 4) + P(X=5) + P(X=6)$

$P(X=5) \Rightarrow \frac{0.0821(2.5)^5}{5!} = 0.0668$

$P(X=6) \Rightarrow \frac{0.0821(2.5)^6}{6!} = 0.0278$

$\therefore P(X \leq 6) = 0.8912 + 0.0668 + 0.0278$

$= 0.9858$

$P(X \geq 6) = 1 - 0.9858$

$= 0.0142$

5. If the probability that an individual suffers a bad reaction due to a certain injections is 0.001.

Determine the probability that out of 2000 individuals (i) Exactly 3 (ii) More than 2 individuals will suffer a bad reaction.

Given $n = 2000$

$p = 0.001$

$\lambda = np = 2000 \times 0.001 = 2$

Poisson probability :

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

(i) probability that exactly 3 individuals suffer a bad reaction.

$$P(X = 3) = \frac{e^{-2}(2)^3}{3!} \qquad [e^{-2} = 0.1353]$$

$$= \frac{0.1353 \times 8}{6}$$

$$= \frac{1.0824}{6} = 0.1804.$$

(ii) probability that more then 2 individuals suffer a bad reaction.

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - P(X=0) + P(X=1) + P(X=2)$$

Now, $P(X=0) \Rightarrow \frac{e^{-2}(2)^0}{0!} = e^{-2} = 0.1353$

$P(X=1) \Rightarrow \frac{e^{-2}(2)^1}{1!} = e^{-1}(2) = 0.2706$

$P(X=2) \Rightarrow \frac{e^{-2}(2)^2}{2!} = 2(e^{-2}) = 0.2706$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$= 0.1353 + 0.2706 + 0.2706$

$= 0.6765$

$P(X > 2) = 1 - 0.6765 = 0.3235$

6. If the masses of 300 students are normally distributed with mean 68 kgs and standard deviation 3 kgs, how many students have masses i, greater than 72kgs, ii, less than or equal to 64 kgs.

Sol

Given Mean (μ) = 68kg

Standard deviation (σ) = 3kg

Total students (N) = 300

w.k.t $Z = \frac{X - \mu}{\sigma}$

i, Students having mass greater then 72kg

$Z = \frac{72 - 68}{3} = \frac{4}{3}$

$= 1.33$

$P(Z > 1.33) = 0.0918$

∴ Number of students $= 300 \times 0.0918$

$= 27.54$

$\approx$ 28 students

ii, Students having mass less than or equal to 64 kg

$Z = \frac{64 - 68}{3} = -\frac{4}{3} = -1.33$

$P(Z < -1.33) = 0.0918$

Number of students = 300 × 0.0918

= 27.54

≈ 28 students

7. A company pay its employees an average wage of ₹ 350 per hour with a standard deviation of ₹ 30. If the wages are approximately normally distributed, Compute the proportion of the workers getting

(i) wages b/w ₹ 325 and ₹ 360 per unit.

(ii) wage is more than ₹ 335, (iii) at least wage is 380 ₹

Sol

Given Mean, $\mu = 350$

Standard deviation, $\sigma = 30$

w.k.t $Z = \frac{X - \mu}{\sigma}$

(i) wages b/w ₹ 325 and ₹ 360

for, 325 ₹

$$Z = \frac{325 - 350}{30} = -0.83$$

for, 360 ₹

$$Z = \frac{360 - 360}{36} = 0.33$$

$P(Z \leq 0.33) = 0.6293$

$P(Z \leq -0.83) = 0.2033$

$P(325 < x < 360) = 0.6293 - 0.2033$

$= 0.4260$

(ii) Wage more than ₹ 335

$$z = \frac{335 - 350}{30} = -0.50$$

$$P(z \leq -0.50) = 0.3085$$

$$P(X > 335) = 1 - 0.3085$$

$$= 0.6915$$

(iii) Wage at least ₹ 380

$$z = \frac{380 - 350}{30} = 1$$

$$P(z \leq 1) = 0.8413$$

$$P(X \geq 380) = 1 - 0.8413 = 0.1587.$$

8. If X is poisson variate such that $3P(X=4) = \frac{1}{2}P(X=2) + P(X=0)$. Find (i) the mean of X (ii) $P(X \leq 2)$

Sol Given $3P(X=4) = \frac{1}{2}P(X=2) + P(X=0)$

Poisson probability

$$P(X=x) = \frac{e^{-\lambda}(\lambda)^x}{x!}$$

$$\Rightarrow 3\left(\frac{e^{-\lambda}(\lambda)^4}{4!}\right) = \frac{1}{2}\left[\frac{e^{-\lambda}(\lambda)^2}{2!}\right] + \frac{e^{-\lambda}(\lambda)^0}{0!}$$

$$3e^{-\lambda}\left[\frac{\lambda^4}{2\times 4}\right] = e^{-\lambda}\left[\frac{\lambda^2}{4} + 1\right]$$

$$\frac{\lambda^4}{8} = \frac{\lambda^2}{4} + 1$$

$$\lambda^4 = 2\lambda^2 + 8$$

let $y = \lambda^2$.

$$y^2 - 2y - 8 = 0$$

$$(y-4)(y+1) = 0$$

Since $y > 0$.

$$y = 4$$

$$\lambda^2 = 4$$

$$\lambda = 2$$

Mean $= 2$

(ii) $P(X \leq 2) \Rightarrow P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$$= e^{-2} + \frac{2e^{-2}}{1!} + \frac{2^2 e^{-2}}{2!}$$

$$= e^{-2}[1+2+2]$$

$$= 5e^{-2}$$

$$P(X \leq 2) = 5(0.1353) = 0.6765$$

9. If x is a normal variance with mean 30 and standard deviation 5. find (i) $P(26 \leq x \leq 40)$.
(ii) $P(x \geq 45)$.

Sol

Given Mean $(\mu) = 30$

Standard deviation $\sigma = 5$ with $z = \frac{x-\mu}{\sigma}$

(i) $P(26 \leq x \leq 40)$

$x = 26 \Rightarrow z = \frac{26-30}{5} = -0.8$

$x = 40 \Rightarrow z = \frac{40-30}{5} = 2$

$$P(26 \leq x \leq 40) = P(-0.8 \leq z \leq 2)$$

$$P(z \leq 2) = 0.9772$$

$$P(z \leq -0.8) = 0.2119$$

$$P(-0.8 \leq z \leq 2) = 0.9772 - 0.2119$$

$$= 0.7653$$

ii) find $P(X \geq 45)$

$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{45 - 30}{5} = 3$$

$$P(X \geq 45) = P(z \geq 3)$$

$$P(z \leq 3) = 0.9987$$

$$P(z \geq 3) = 1 - 0.9987$$

$$= 0.0013$$

10. The length of time a person speaks over phone follows exponential distribution with mean 6. what is the probability that the person will talk for

i) more than 8 minutes.

ii) between 4 and 8 minutes.

Sol Given Mean = 6 minutes

$$\lambda = \frac{1}{\text{Mean}} = \frac{1}{6}$$

Probability density function

$$f(x) = \lambda e^{-\lambda x}, x \geq 0$$

$$f(x) = \frac{1}{6} e^{-x/6}$$

i) probability for more than 8 minutes.

for an exponential distribution,

$$P(X > a) = e^{-\lambda a}$$

Substitute $\lambda = 1/6$ and $a = 8$.

$$P(X > 8) = e^{-8/6} = e^{-4/3}$$

$$= 0.2636$$

(ii) probability that the person talks b/w 4 and 8 minutes.

$$P(4 < x < 8) = P(X > 4) - P(X > 8)$$

$$P(X > 4) = e^{-4/6} = e^{-2/3}$$

$$= 0.5134$$

$$P(X > 8) = 0.2636$$

$$P(4 < X < 8) = 0.5134 - 0.2636$$

$$= 0.2498$$